Ling 285
Human Language and Technology
Lecture 7

Dynamical Systems 5

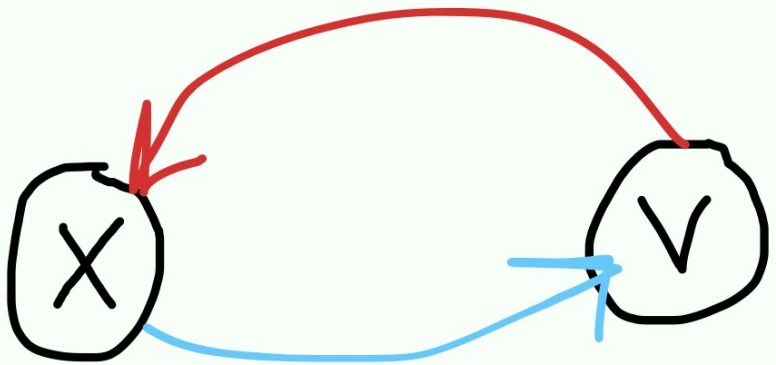
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Harmonic Oscillator

$$\dot{x} = v$$

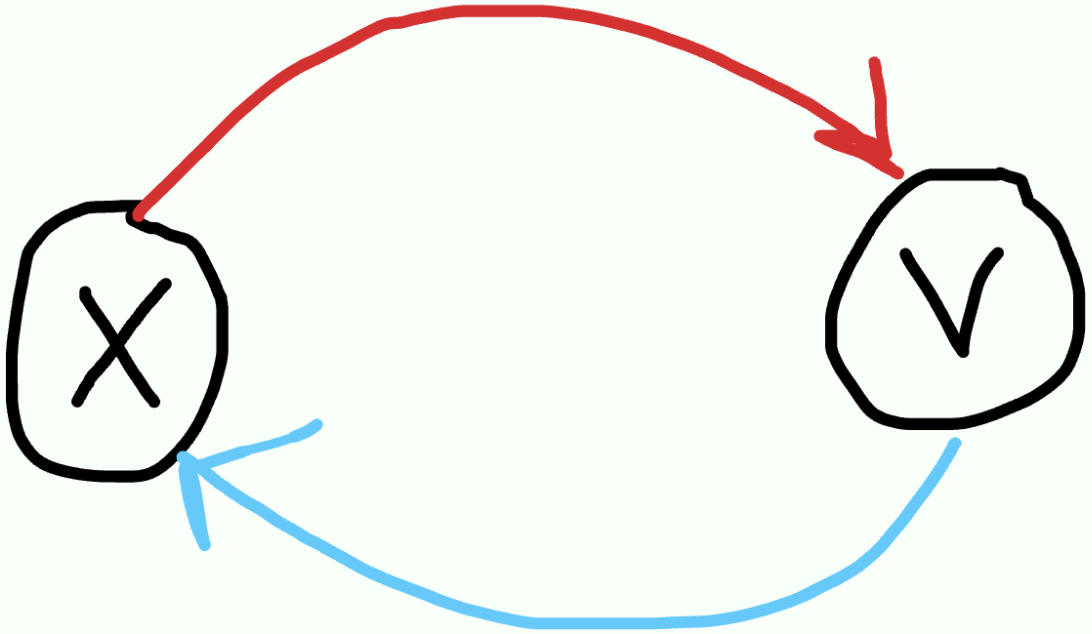


$$\dot{v} = -x$$

$\dot{x}$ means: increase or decrease in x with time

$\dot{v}$ means: increase or decrease in v with time

Oscillation/Vibration Dynamical System



Vibration needs two physical effects

- One effect is called a (viscous) spring. It’s the springiness in many things around you. (I will ignore “viscous”). It has a property we call stiffness.
- The other effect is called mass.
- We will now discuss each of these to understand what vibration is, and remmebr how important vibration is:
  - It’s what you feel when you hit a bump in your car
  - It’s music
  - It’s boom and bust in economics
  - It’s oscillation in SGD
  - It’s the motion of the air in your vocal tract and the motion of your vocal folds, when you speak

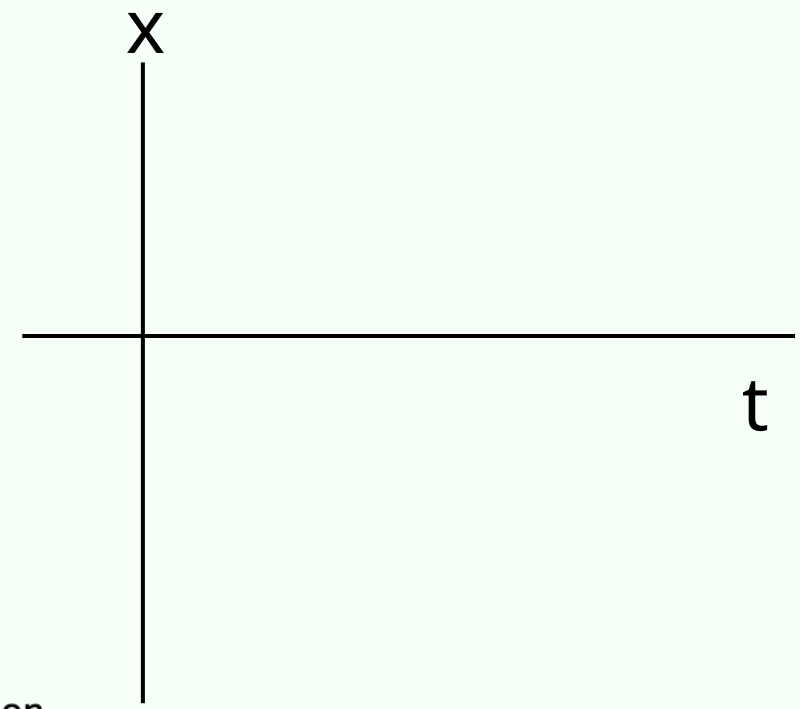
Slice of air in your vocal tract

High pressure means air very close

Low pressure means air very far apart

Viscous spring, Negative/Balancing feedback

- $\dot{x} = -\frac{1}{3}x$
- If x is positive, will $\dot{x}$ rise or fall?
- If x is positive, $\dot{x}$ will fall?
- If x is negative, will $\dot{x}$ rise or fall?
- If x is nagative, $\dot{x}$ will rise?
- x initial condition = 1. Will x rise or fall?
- Fall by 1/3 of its current size
- Now what will happen? Again and Again
- When x is smaller, $\dot{x}$, its speed of falling is smaller!
- Graphically, we're seeing the slope get smaller and smaller negative
- What if the initial condition is negative?
- As you've learnt from Meadows, these are systems that have a natural resting condition, and when their position is at a discrepancy with their natural position, they will *feedback* on themselves $\dot{x} = -\frac{1}{3}x$ to cancel the discrepancy
- These are systems that are in motion *only in order to achieve their equilibrium*
- -1/3 is what we call stiffness. Higher? Lower?

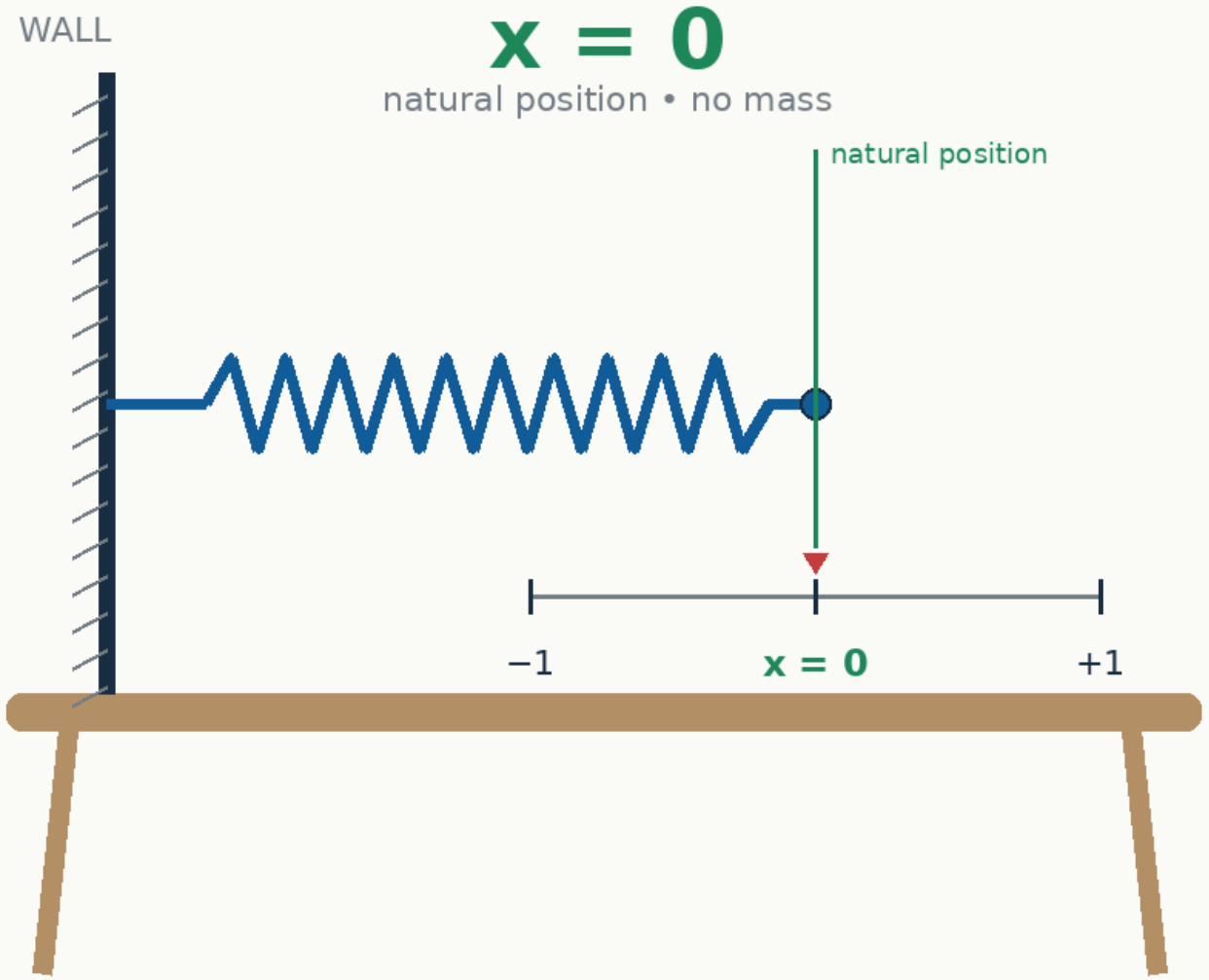


Now let’s talk about Mass

- Hmmmm…
- If a mass is moving, it just keeps on moving
- Galileo, Descartes, Newton: A mass in motion stays in motion.



A spring stops at a place. A mass stops after a process.



Displacement x is measured from the spring’s natural position.

There’s a cosine in your mouth!

Vowels

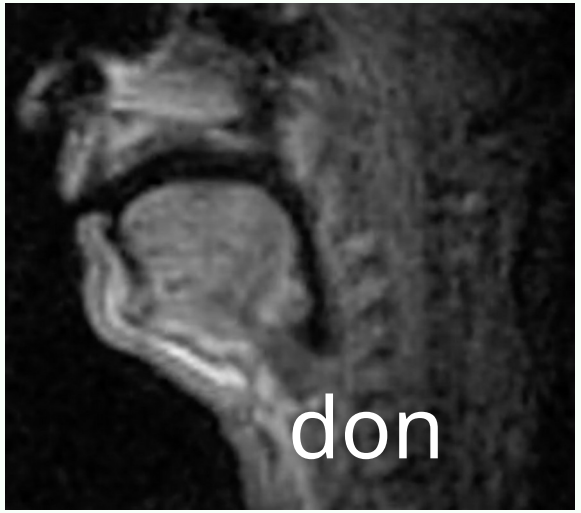
High





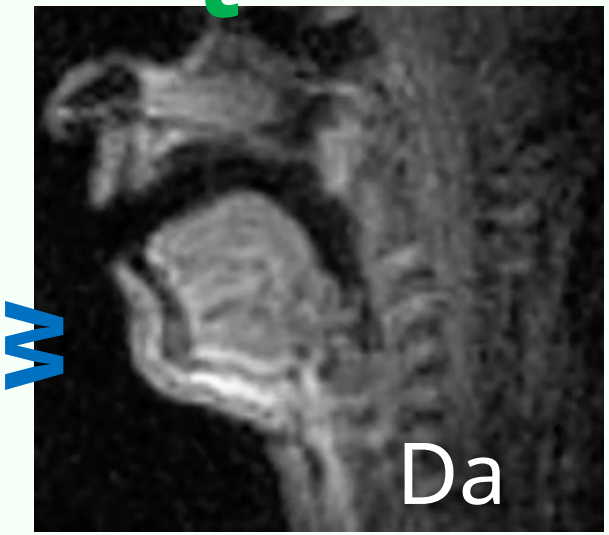
Front

Back



Central

Low





Perturbation Theory

- If we increase the mass m of a mass-spring system, will the frequency increase or decrease?
- Frequency will Decrease
- If we increase the stiffness k?
- Frquency will Increase
- Perturbation Theory:
  - m ↑ f↓
  - k ↑ f↑

Hearing
Explained!

Georg von Bekesy Nobel prize in 1961

Fundamental Learning Tradeoff

- $\gamma \uparrow$ Speed ↑ Accuracy ↓
- $\gamma \downarrow$ Speed ↓ Accuracy ↑
- ☹️
- Thousands of people for generations make their living by finding better way to *optimize* so that
  - Speed ↑
  - Accuracy ↑

